

Haardhik Ratnakar Kunder

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Education

TU Dresden, Dresden 2025 – Present
MSc in Computational Modeling and Simulation Applied AI track (2nd Semester)

University of Mumbai, Mumbai 2021 – 2025
Bachelor of Technology in Computer Engineering, Honors in Intelligent Computing

Research

Latent Communication for Multi-Trajectory Data Analysis Agents [GitHub](#) Jun 2026

- Tested whether code-executing data-analysis agents can replace natural-language handoffs with shared hidden-state memory, comparing **single-agent**, **text based**, and **latent planner-coder-critic** modes.
- Showed latent coordination reduced decoded coordination tokens to **0** while text coordination grew from **979 to 1768 to 2677 tokens** across horizons, with short/medium latent pass rates of **1.00/1.00** versus text **0.93/0.87**.

MedFiber: Synthetic 3D Biomedical Fibre Segmentation [GitHub](#) May 2025 – Ongoing

- Developed an end-to-end **synthetic 3D imaging pipeline** that generates realistic **3D volumetric environments** with voxel-level ground-truth labels, directly addressing the labeled data scarcity that prevents training production-grade 3D segmentation systems.
- Simulated dense, intertwined, non-intersecting structures using **Blender physics simulation** and converted them into grayscale microscopy-style volumes, producing contact regions and boundary masks that replicate real scanning conditions.
- Validated predictions against known ground truth using **Dice, ARI, IoU, object F1, and variation of information** metrics, and built 3D inspection tooling that cut debugging time for algorithm iteration on split/merge failures.

Arthantar: Contextual Translation System (Bachelor Thesis) [GitHub](#) Jan 2025

- Investigated whether smaller, cheaper models could match larger ones purely through improved surrounding architecture, targeting reduced inference cost without any retraining or fine-tuning.
- Constructed **dynamic knowledge graphs** using **LLMGraphTransformer** with **FastCoref** for coreference resolution, testing across **Mistral, LLaMA3, and Gemma** to supply each model with explicit relational structure.
- Used **spaCy** dependency parsing and NER to build fallback graphs, enabling the system to consistently outperform much larger LLMs on **gender-aware, cross-lingual translation** benchmarks.

Federated Learning & Explainable AI for Tumor Prediction [GitHub](#) Jun 2024 – Aug 2024

- Achieved **96.07%** brain tumor detection accuracy using a federated **ResNet50 + FFT** architecture trained across decentralized nodes, improving over the centralized CNN baseline (**86.46%**) by **9.6 percentage points** while preserving patient data privacy in compliance with **HIPAA** and **GDPR**.
- Integrated **SHAP** and **LIME** to surface which image regions drove model decisions, transforming opaque CNN predictions into clinically inspectable outputs aligned with known tumor features.

Projects

Dresden-Live: 3D Geospatial Transit Engine [GitHub](#) Mar 2026

- Architected a **3D live transit dashboard** using **Three.js** and **Mapbox GL JS** to visualize live VVO telemetry with **sub-second updates** across the Dresden transit network.
- Implemented resilient **async polling and buffering** to maintain interface responsiveness under API jitter, sustaining smooth rendering during high-frequency data bursts.

Spending Rail TinyFish Accelerator [Link](#) Apr 2026 – Present

- Selected for **Phase 2 of the TinyFish Accelerator**, a 9-week program backed by Mango Capital with a **\$2M seed funding pool**.
- Architected a **banking layer for autonomous agents** that provisions each agent with a scoped wallet, enforces spend limits and risk-aware credit lines, and routes all purchases through a **single reconciliation layer**, eliminating manual spend oversight across multi-agent deployments.

Automated Book Publication [GitHub](#) May 2025 – Jun 2025

- Designed a modular framework that automated book chapter transformation using **Playwright** for scraping and **LLaMA3** and **Deepseek-R1** for rewriting, reducing manual editing overhead for the human-in-the-loop **Streamlit** review stage.

- Integrated **FAISS** and **SentenceTransformers** for semantic search alongside a custom **RL-driven feedback loop** that ranked chapter versions by narrative quality and stylistic attributes, enabling consistent selection of highest-quality rewrites.

Publications

Exploring Deep Learning Techniques for Topic Modeling of Natural Disaster Tweets 2025
Springer Nature [Link](#)

- Investigated how people think and communicate under acute stress by analyzing tweet patterns during hurricanes, wildfires, and floods, treating social media as a lens into collective cognitive and emotional states during crisis.
- Evaluated four topic modeling approaches (LSA, LDA, BERT, and LLaMA3 with BERTopic) to identify which methods best recover latent behavioral structure hidden in brief, unstructured text produced under high-pressure conditions.
- Engineered a custom **LLaMA3 + BERTopic** pipeline with **UMAP** and **HDBSCAN** clustering that outperformed all baselines in distinguishing nuanced theme clusters.

Hybrid Predictive Modeling for F1 Races: Random Forest and Graph Neural Networks 2026
Springer Nature [Link](#)

- Designed a sequential **RF-to-GNN** pipeline that achieved a **macro F1 of 0.81**, surpassing standalone GNN (0.72) and logistic regression (0.67) baselines by modeling relational and temporal dynamics across drivers, teams, and race conditions.
- Applied **SMOTE** and **GridSearch** hyperparameter tuning to resolve class imbalance, boosting Random Forest recall from **0.24 to 0.50** and F1 from **0.32 to 0.48** across sparse, high-dimensional race event data.

Emotion-Aware Indian Sign Language Recognition: A Multimodal Approach 2024
Educational Administration: Theory and Practice [Link](#)

- Constructed a real-time multimodal system combining **CNN (79.22%** emotion detection accuracy) and **LSTM (98.12%** gesture recognition accuracy) to simultaneously decode sign sequences and infer affective state from facial dynamics.
- Synthesized gesture and expression streams through correlation analysis and integrated **Google Text-to-Speech** output across **5 Indian regional languages**.

Experience

Web Development Intern, Concetto Media Apr 2023 – Sep 2023
Aurangabad, India

- Developed a full-stack website from scratch for an educational institution serving **15,000+ students**, delivering the complete product independently end to end.
- Optimized site architecture and on-page **SEO**, improving search engine visibility and measurably increasing organic traffic for the institution by **40 percent**.

Skills

Programming Languages: Python (Advanced), C/C++, JavaScript, MATLAB

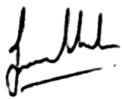
Robotics & Simulation: Blender (Physics Simulation), Open3D, 3D Scanning

Machine Learning & AI: PyTorch, TensorFlow, LLMs (Advanced), GNNs, OpenCV, LangChain, Google ADK

Web Dev & Databases: ReactJS, Node.js, Flask, Three.js, Tailwind CSS, MongoDB, MySQL, Selenium

Tools & Platforms: Docker, AWS, Git, Codex, Claude Code, Kubernetes

Languages: English (C1), German (A1)



Dresden, June 23rd